Three-class classification of Motor Imagery EEG data including "rest state" using Filter-Bank multi-class Common Spatial Pattern

T. SHIRATORI, H. TSUBAKIDA, A. ISHIYAMA
School of Advanced Science and Engineering.
Waseda Univ.
Tokyo, Japan.

Y. ONO
Dept. of Elec. and Bioinfomatics, Sch. of Sci. and Tech.
Meiji Univ
Kanagawa, Japan.

Abstract— **Our purpose is to develop the 3-class Brain Machine Interface (BMI) incorporating the classification of resting state using Electroencephalography (EEG). Conventionally the most of BMI systems only accept EEG data when a subject performs some kind of task, such as motor imagery and gaze at visual stimuli. However, performing task causes fatigue of the subject. It is therefore important to develop classification algorithm for BMI system that utilizes rest state-EEG as one of the classes. The 3 classes we defined in this experiment were: (1) motor imagery of moving right hand; (2) motor imagery of moving left hand; and (3) rest state. And, we also measured EEG in an actual moving task (finger tapping) to ascertain validity of algorithm. We extracted feature vector using Finite Impulse Response (FIR) digital filter Filter Bank and multi-class Common Spatial Filter (mCSP) from EEG data, selected the feature by Mutual Information (MI), and made three 3-class classifiers using Random Forest (RF). The mean classification rate was 56.7±4.43% at motor imagery task and 88.7±4.54% at actual finger tapping task. And we compared the time required to extract features and compute classifiers with those of other methods. Our method is effective to some extent. (1) parameter selection time was better than choosing single band-pass filter that best discriminate classes among possible options of frequency bands; and (2) accuracy rate was better than our previous method using majority vote.**

Keywords-EEG classification; motor imagery;reststate;multi-class common spatial pattern;filter bank common spatial pattern ;mutual information; random forest;

Introduction

Brain machine interface (BMI) leads to develop a technique to support independence of physically challenging person and that to improve quality of everyday life of healthy person. While there are several kinds of measurement procedure of the brain activity, we used Electroencephalography (EEG) which has high temporal resolution and portability. Accurate classification of EEG data is necessary to realize feasible BMI system. However, EEG has low spatial resolution and signal contains many noises. Due to the noises, classification of EEG requires preprocessing, extracting features and sometimes classifier. We studied 3-class BMI incorporating the classification of resting state in consideration of fatigue of the subject. The 3 classes we defined in this experiment were (1) motor imagery of moving right hand;(2) motor imagery of moving left hand; and (3) rest state. Our proposed method uses multi-class common spatial patterns (mCSP) [1], Filter Bank using feature selection by Mutual Information (MI), and Random Forest (RF) [2][3]. RF is known as effective classifier for EEG-data [4].Usually Common Spatial Patterns (CSP) [5] is used for 2 classes EEG discrimination. But CSP can be extended to multi class data by Joint Approximate Diagonalization (JAD) [6][7].This CSP is called mCSP. Also there is a problem that CSP requires the frequency band of the band pass filter as a hyper parameter. Filter Bank Common Spatial Pattern (FBCSP) [8] is known as a method to solve this problem. We attempt to apply FBCSP for mCSP. We made 3-class classifiers by using RF. Our proposed method was better than using single band-pass filter method regarding parameter selection time in motor imagery task. In finger tapping task, we achieved better classification rate than our previous method using majority voting.

I. EXPERIMENTS

A. data acquisition

Eight healthy young subjects (20-25 years old) participated in the study (subject A, B, C, D, E, F, G and H). This experiment was approved from the Waseda University Ethical Review Board. In accordance with the international 10-20 system, we measured their EEG signals from 15 electrodes distributed over the scalp (Fp1, Fp2, F3, Fz, F4, T3, C3, Cz, C4, T4, P3, Pz, P4, O1, O2), sampled at 125 Hz. The procedure of experiment is illustrated in Figure 1. Subjects looked at a computer screen in front of them during the whole experiment. The screen alternately showed an instruction (4s) and a fixation point (3s) for 60 times. An instruction of single state among three states (1. motor imagery of moving right hand, 2.motor imagery of moving left hand, 3. rest state) was displayed at the beginning of instruction display for 1s, and count-down was displayed for the following 3s. After the count-down subject performed the instructed state for 3s. To reduce artifacts, subject was encouraged not to blink and move his eyes. Subject performed three kinds of states each for 20 times. We also measured actual moving task (finger tapping) to ascertain validity of the developed algorithm.

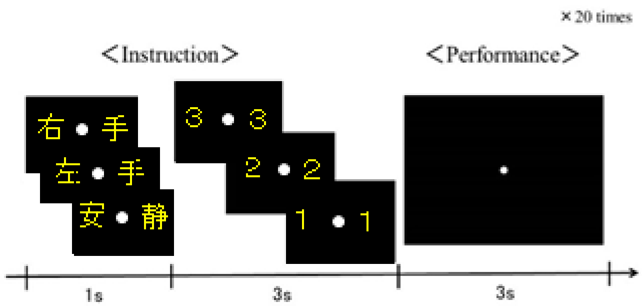

Figure1. Experiment procedure (安静 means rest state 右手 means right hand 左手 means left hand. To clarity characters were written in yellow and bigger than the actual experiment. They were shown in blue and their size was smaller in the actual experiment to avoid visual evoked responses.)

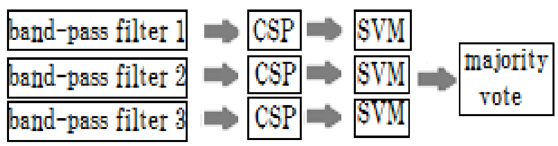

Figure2. Previous method

Algorithm

Our previous method is illustrated in Figure2.

This method uses three band-pass filters, CSP and Support Vector Machine (SVM). Majority voting among the classified results from 2-class classifiers decided the final output of the 3-class classifier.

Common Spatial Pattern(CSP)

CSP is a two-class classifier of multichannel datasets. CSP filter maximizes the ratio of the variance of the data in one class while minimize that in another class. Band-pass filtered EEG-data is represented as:

$$\boldsymbol{E}_j \in \boldsymbol{R}^{m \times k} \tag{1}$$

where, j denotes the number of trials, m denotes the number of channels, k denotes the number of data from each trial. We can estimate of the covariance matrix of $\boldsymbol{E}_j$:

$$\boldsymbol{\Sigma}^{C} = \sum_{j \in C^i} \frac{\boldsymbol{E}_j \boldsymbol{E}_j^T}{trace(\boldsymbol{E}_j \boldsymbol{E}_j^T)}, \quad (i \in [1,2]) \tag{2}$$

CSP filter $\boldsymbol{W}$ is given by simultaneous diagonalization between $\boldsymbol{\Sigma}^1$and $\boldsymbol{\Sigma}^2$:

$$\boldsymbol{W}^T \boldsymbol{\Sigma}^{(1)} \boldsymbol{W} = \boldsymbol{\Gamma}^{(1)}, \boldsymbol{W}^T \boldsymbol{\Sigma}^{(2)} \boldsymbol{W} = \boldsymbol{\Gamma}^{(2)} \tag{3}$$

$$\boldsymbol{\Gamma}^{(1)} + \boldsymbol{\Gamma}^{(2)} = \boldsymbol{I} \tag{4}$$

$\boldsymbol{\Gamma}^{(1)}$and $\boldsymbol{\Gamma}^{(2)}$are diagonal matlix. CSP filter $\boldsymbol{W}$ coressponds to eigenvectors:

$$\boldsymbol{W} = [\boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_m] \tag{5}$$

The eigenvectors can be given by solving the genelized eigenvalue problem:

$$\boldsymbol{\Sigma}^{(1)} \boldsymbol{\omega} = \lambda \boldsymbol{\Sigma}^{(2)} \boldsymbol{\omega} \tag{6}$$

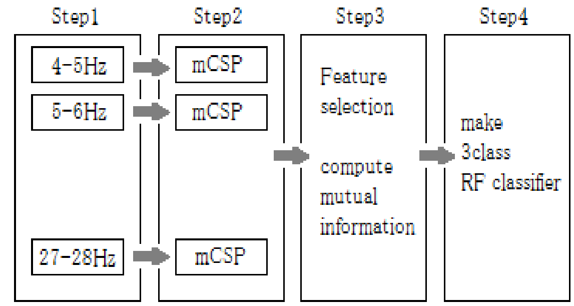

Figure3.Algolithm step

Our proposed method is consisted of the following 4 steps as shown in Figure3.

Step1. Make band-pass filtered EEG-data using FIR digital filters of different frequency width-parameter.

Step2 Calculate mCSP filter from band-pass filtered EEG-data, and extract feature vectors given by variance.

Step3 Compute MI between features and labels and select input features.

Step4 Make a 3-class classifiers using Random Forest (RF).

Multiclass Common Spatial Pattern

CSP is usually applicable to 2 class classification problems, because CSP filter needs simultaneous diagonalization [9]. In case of simultaneous diagonalization of the more than 2 matrix, we can use JAD to solve this problem. Covariance matrix $\boldsymbol{\Sigma}^{(c^i)} (i \in [1,2,3])$ is given as in (2). Composite covariance matrix can then be given as:

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(1)} + \boldsymbol{\Sigma}^{(2)} + \boldsymbol{\Sigma}^{(3)} \tag{7}$$

To calcurate whitening matrix, first diagonalize composite matlix as follows:

$$\boldsymbol{\Sigma} = \boldsymbol{U}_0 \boldsymbol{R}_0 \boldsymbol{U}_0^T \tag{8}$$

U_0 denotes matrix of eigenvectors, R_0 denotes diagonal matrix of eigenvalues. Whitening matrix is given by:

$$\boldsymbol{P} = \boldsymbol{R}_0^{-\frac{1}{2}} \boldsymbol{U}_0^T \tag{9}$$

Covariance matlix is transformed by whitening matlix *P:*

$$\boldsymbol{S}^{(i)} = \boldsymbol{P} \boldsymbol{\Sigma}^{(i)} \boldsymbol{P}^T \ (i \in [1,2,3]) \tag{10}$$

Approximate simultaneous diagonalization is given by V which can be calculate by JAD:

$$\boldsymbol{V}^T \boldsymbol{S}^{(i)} \boldsymbol{V} = \boldsymbol{\Gamma}^{(i)} (i \in [1,2,3]) \tag{11}$$

We can compute mCSP filter:

$$\boldsymbol{W} = \boldsymbol{V}^T \boldsymbol{P} \tag{12}$$

Spatial filtered signal $\boldsymbol{Z}$ of EEG data $\boldsymbol{E}$ is given as:

$$\boldsymbol{Z} = \boldsymbol{W} \boldsymbol{E} \tag{13}$$

feature is extracted by calcurate normalized variance of spatial filtered signal Z as:

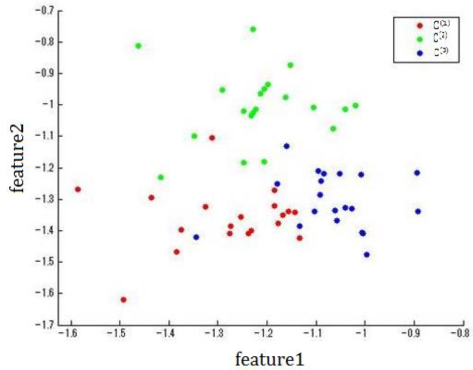

Figure4. Satter diagram in the 2 dimentional features of representative data in the experiment

$$f_p = \log\left(\frac{var(\boldsymbol{Z}_p)}{\sum_1^m var(\boldsymbol{Z}_p)}\right) \quad p \in \{1 \dots m\} \qquad (14)$$

where *m* denotes number of chanel. Figure4 shows an example scatter diagram in the 2 dimentional features.

Feature selection by Mutual Information

We can get features comprised of many dimentions at step2. But all dimensions are not necessarily the good feature. And the more dimensions require the more time-consuming calculation. We therefore attempt to decrease dimension of features. There are many index to select feature [9]. We choose MI of features and class labels. MI is given by:

$$I(f;l) = \sum_f \sum_l p(f,l) \log_2\left(\frac{p(f,l)}{p(f)p(l)}\right) \qquad (15)$$

where *f* denotes feature*, l* denotes class label. Joint probability can be estimated by conditional probability:

$$p(f,l) = p(f|l)p(l) \qquad (16)$$

$p(f|l)$ was calculated by assuming a Gaussian distribution. We choose features which have *a*% of maximum mutual information.We tried *a* between 20 and 80.

Random Forest

RF classifier is one of the ensemble learning algorithms which consists of many binary decision trees. An example of a binary decision tree is illustrated in Figure6. It has some nodes and leaves which output final result. Node has split function which decides one of the two child nodes. Split function is represented as:

$$h(\boldsymbol{f}, \boldsymbol{\theta_k}) \in (0,1) \qquad (17)$$

Here, f denotes feature vector reached to the node, θ_k denotes parameter of split functions and *k* is the number of the node. When feature vector was given to the tree, it reaches to the first node. In the node, split function determines which child node to be selected. Repeating this decision, finally it reaches to one of the leaves and the tree gives output decision. One decision tree gives one class C^i. Although a single decision tree may not be a good classifier, combining the

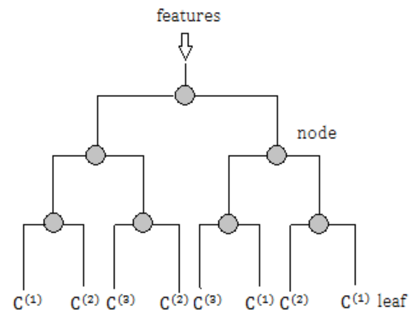

Figure5.Abinaly decision tree

decision of many trees can compose reliable classifier by compensating each other [4]. Features given to each tree is different. $f_m \in f$ denotes input to the m-th tree. RF classifier decides the class by majority vote of many decision trees.

II. RESULTS

Table1 and Table 2 show classification results of motor imagery and finger tapping tasks, respectively. Method 1 shows the result when data was classified using a 2-class classifiers with Gaussian kernel SVM and majority vote. We tried the 7 different band-pass filters (2 alpha wave, 3 beta wave, and 2 alpha and beta wave bandwidths). Method 2 shows the result when data was classified using only single band-pass filter and mCSP. We tried the 400 different band-pass filters between 1Hz ~ 40Hz and choose the best filter. Method 3 shows the result obtained by FBmCSP. We use 24 band-pass filters among 4Hz and 28Hz. Here, n_f denotes the number of features selected by MI. Parameter selection time indicates the time spent to choose parameters. Method 1 and Method 2 have width parameter of band-pass filter. Method 3 has threshold *a* of MI. Width parameter tended to be chosen in the beta frequency band. *a* tend to be chosen between 20~60. Computing time expresses the time that a calculation using best parameters for the one subject takes. We evaluated the classification rate using leave-one-out cross-validation.

TABLE I. THE CLASSIFICATION RATE AND TIME USING METHOD 1,2,3 FOR MOTOR IMAGERY TASK

	Method 1	Method 2	Method 3	
subject	rate	rate	Rate	n_f
A	63.3%	50%	55%	263.4
B	65%	66.7%	63.3%	27.0
C	55%	60%	46.7%	29.9
D	58.3%	65%	58.3%	69.9
E	40%	55%	55%	66.5
F	63.3%	56.7%	56.7%	64.2
G	48.3%	53.3%	56.7%	94.9
H	45%	55%	61.7%	119.4
Ave	54.8%	57.7%	56.7%	91.9
Parameter selection time	603sec	10731sec	864sec	
Computing time	67sec	24sec	587sec	

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TABLE II THE CLASSIFICATION RATE AND TIME USING METHOD 1,2,3 FOR FINGER TAPPING TASK

	Method 1	Method 2	Method 3	
subject	rate	rate	rate	subject
A	78.3%	88.3%	93.3%	74.0
B	65%	71.7%	81.7%	114.9
C	81.7%	91.7%	93.3%	55.5
D	73.3%	91.7%	91.7%	19.2
E	70%	80%	90%	48.1
F	86.7%	93.3%	93.3%	33.3
G	71.7%	81.7%	85%	22.5
H	56.7%	70%	81.7%	74.6
Ave	72.9%	83.5%	88.7%	55.3
Select Parameter time	603sec	10731sec	865sec	
Computing time	67sec	24sec	596sec	

III. DISCUSSION

Table1 shows that there are not much difference in the classification rate among the three algorithms (method of Bonferroni for multiple comparison. p=0.98 (Method1 and Method2), p=1.00 (Method2 and Method3), p=1.00(Method1and Method3)). However, the computation times are greatly different. It is necessary for method 1 to make three 2-class classifier and band-pass filter. Therefore the computing time of Method 1 is worse than Method 2. Method 2 spends less computing time, because it requires only one 3-class classifier and band-pass filter. On the other hand, its Parameter selection time is the longest. It is due to the many trials required to determine the best band-pass filter. Method 3 solves these problems. It does not need to compute many band-pass filters. MI can reduce the time for classifier to learn. But the requirement of relatively more band-pass filters made the computing time the worst. We need to further investigate the appropriate number of band-pass filters to take balance between the accuracy and the computing time. Table2 shows that Method 3 is the most effective method regarding as classification rate (p=0.000054 (Method1 andMethod2), p=0.0240 (Method2 and Method3), p=0.000051 (Method2 and Method3); Bonferroni correction for multiple comparison.). This suggests that the applying Method 3 for task which is easy to classify has a more potent. The subjects of this experiment were never been trained. It is reported that subject who trained motor image task can achieve good score [9]. Method 3 may be suitable by the subject who trained. Because Method 3 is best method task which is easy to classify. Figure 6 shows relationship of MI and the number of features. The maximum MI of finger tapping task is mostly higher than that of motor imagery task. It suggests that classification rate and the maximum MI have a strong relationship. As much as maximum MI higher, classification rate is better.

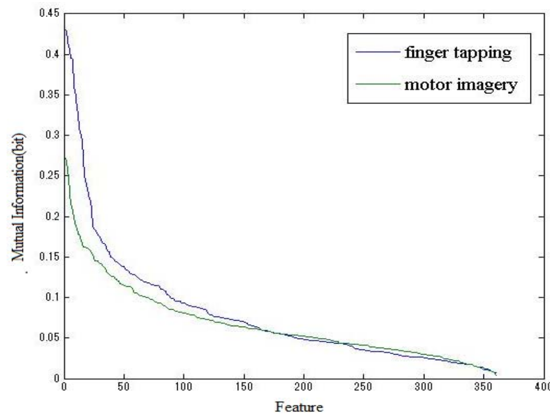

Figure6. mutual information

IV. CONCLUSION

We proposed a new method for classifying motor Imagery EEG data including "rest state". We achieved an averaged classification late 56.7±4.4% in motor imagery task, and 88.7±4.54% in finger tapping task. This result suggested that our proposed method have better classification rate in three-class conditions compared to the majority voting algorithm based on two-class classifiers. As a future task, we intend to compare the proposed method with other feature selection methods. Decreasing computing time would also be necessary.

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